РОССИЙСКИЙ УНИВЕРСИТЕТ ДРУЖБЫ НАРОДОВ

Факультет физико-математических и естественных наук

ОТЧЁТ

ПО ЛАБОРАТОРНОЙ РАБОТЕ №14

дисциплина: Администрирование сетевых подсистем

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МОСКВА

2025 г.

Цель работы:

Целью данной работы является приобретение навыков настройки доступа групп пользователей к общим ресурсам по протоколу SMB.

Выполнение работы:

```
root@server:~
agrant@server ~]$ dnf -y install samba samba-client cifs-utils
ror: This command has to be run with superuser privileges (under the root user on mos
systems).
agrant@server ~]$ sudo -i
oot@server ~]# dnf -y install samba samba-client cifs-utils
cky Linux 9 - BaseOS                          5.6 kB/s | 4.3 kB     00:00
cky Linux 9 - AppStream                       5.7 kB/s | 4.8 kB     00:00
cky Linux 9 - Extras                          7.5 kB/s | 3.1 kB     00:00
pendencies resolved.
================================================================================
ackage                    Arch       Version              Repository     Size
================================================================================
stalling:
ifs-utils                 x86_64     7.2-1.el9            baseos        113 k
amba                      x86_64     4.22.4-12.el9_7      baseos        930 k
amba-client               x86_64     4.22.4-12.el9_7      appstream     760 k
stalling dependencies:
ibnetapi                  x86_64     4.22.4-12.el9_7      baseos        142 k
amba-common-tools         x86_64     4.22.4-12.el9_7      baseos        483 k
amba-dcerpc               x86_64     4.22.4-12.el9_7      baseos        717 k
amba-ldb-ldap-modules     x86_64     4.22.4-12.el9_7      baseos         34 k
amba-libs                 x86_64     4.22.4-12.el9_7      baseos        123 k

ansaction Summary
================================================================================
stall  8 Packages

tal download size: 3.2 M
stalled size: 12 M
```

smb.conf
/etc/samba

```
# See smb.conf.example for a more detailed config file or
# read the smb.conf manpage.
# Run 'testparm' to verify the config is correct after
# you modified it.
#
# Note:
# SMB1 is disabled by default. This means clients without support for SMB2 or
# SMB3 are no longer able to connect to smbd (by default).

[global]
        workgroup = BAGRANT-NET

        security = user

        passdb backend = tdbsam

        printing = cups
        printcap name = cups
        load printers = yes
        cups options = raw

[homes]
        comment = Home Directories
        valid users = %S, %D%w%S
        browseable = No
        read only = No
        inherit acls = Yes

[printers]
        comment = All Printers
        path = /var/tmp
        printable = Yes
        create mask = 0600
        browseable = No

[print$]
        comment = Printer Drivers
        path = /var/lib/samba/drivers
```

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```
Complete!
[root@server ~]# groupadd -g 1010 sambagroup
[root@server ~]# usermod -aG sambagroup user
[root@server ~]# mkdir -p /srv/sambashare
[root@server ~]# gedit /etc/samba/smb.conf

(gedit:9023): dconf-WARNING **: 15:20:19.112: failed to commit changes to dconf: Failed
 to execute child process “dbus-launch” (No such file or directory)

(gedit:9023): dconf-WARNING **: 15:20:19.121: failed to commit changes to dconf: Failed
 to execute child process “dbus-launch” (No such file or directory)

(gedit:9023): dconf-WARNING **: 15:20:19.620: failed to commit changes to dconf: Failed
 to execute child process “dbus-launch” (No such file or directory)

(gedit:9023): dconf-WARNING **: 15:20:19.620: failed to commit changes to dconf: Failed
 to execute child process “dbus-launch” (No such file or directory)
```

```
35 [print$]
36         comment = Printer Drivers
37         path = /var/lib/samba/drivers
38         # printadmin is a local group
39         write list = printadmin root
40         force group = printadmin
41         create mask = 0664
42         directory mask = 0775
43 [sambashare]
44         comment = My Samba Share
45         path = /srv/sambashare
46         write list = @sambagroup
```

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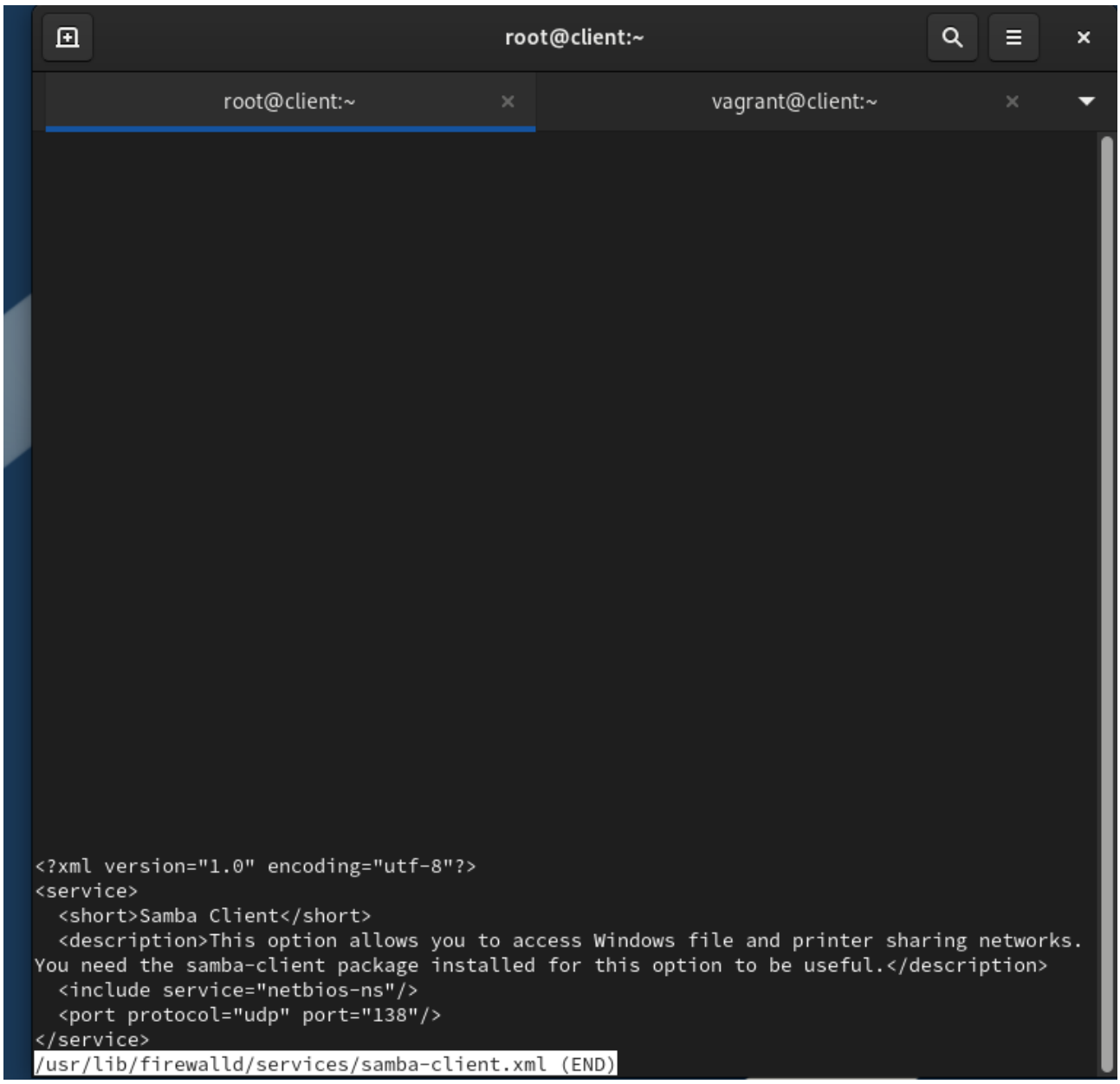
```
<?xml version="1.0" encoding="utf-8"?>
<service>
  <short>Samba</short>
  <description>This option allows you to access and participate in Windows file and pri
nter sharing networks. You need the samba package installed for this option to be usefu
l.</description>
  <include service="samba-client"/>
  <port protocol="tcp" port="139"/>
  <port protocol="tcp" port="445"/>
</service>
/usr/lib/firewalld/services/samba.xml (END)
```

root@server:/srv

root@server:~ root@server:/srv

```
[vagrant@server ~]$ sudo-i
bash: sudo-i: command not found...
[vagrant@server ~]$ sudo -i
[root@server ~]# firewall-cmd --add-service=samba
success
[root@server ~]# firewall-cmd --add-service=samba --permanent
success
[root@server ~]# firewall-cmd --reload
success
[root@server ~]# chgrp sambagroup /srv/sambashare
[root@server ~]# chmod g=rwx /srv/sambashare
[root@server ~]# cd /srv
[root@server srv]# ls -Z
unconfined_u:object_r:nfs_t:s0 nfs  unconfined_u:object_r:var_t:s0 sambashare
[root@server srv]# semanage fcontext -a -t samba_share_t "/srv/sambashare(/.*)?"
[root@server srv]# restorecon -vR /srv/sambashare
Relabeled /srv/sambashare from unconfined_u:object_r:var_t:s0 to unconfined_u:object_r:
samba_share_t:s0
[root@server srv]#
```

```
root@client:~
[vagrant@client ~]$ sudo -i
[root@client ~]# dnf -y install samba-client cifs-utils
Rocky Linux 9 - BaseOS                          8.6 kB/s | 4.3 kB     00:00
Rocky Linux 9 - BaseOS                          6.5 MB/s |  16 MB     00:02
Rocky Linux 9 - AppStream                        15 kB/s | 4.8 kB     00:00
Rocky Linux 9 - AppStream                       4.4 MB/s |  17 MB     00:03
Rocky Linux 9 - Extras                          7.3 kB/s | 3.1 kB     00:00
Rocky Linux 9 - Extras                           33 kB/s |  17 kB     00:00
Dependencies resolved.
================================================================================
 Package                Arch        Version                Repository      Size
================================================================================
Installing:
 cifs-utils             x86_64      7.2-1.el9              baseos         113 k
 samba-client           x86_64      4.22.4-12.el9_7        appstream      760 k
Upgrading:
 libipa_hbac            x86_64      2.9.7-4.el9_7.1        baseos          33 k
 libldb                 x86_64      4.22.4-12.el9_7        baseos         180 k
 libsmbclient           x86_64      4.22.4-12.el9_7        baseos          73 k
 libsss_certmap         x86_64      2.9.7-4.el9_7.1        baseos          88 k
 libsss_idmap           x86_64      2.9.7-4.el9_7.1        baseos          39 k
 libsss_nss_idmap       x86_64      2.9.7-4.el9_7.1        baseos          43 k
 libsss_sudo            x86_64      2.9.7-4.el9_7.1        baseos          33 k
 libtalloc              x86_64      2.4.3-1.el9            baseos          33 k
 libtdb                 x86_64      1.4.13-1.el9           baseos          53 k
 libtevent              x86_64      0.16.2-1.el9           baseos          50 k
 libwbclient            x86_64      4.22.4-12.el9_7        baseos          42 k
 samba-client-libs      x86_64      4.22.4-12.el9_7        baseos         5.3 M
 samba-common           noarch      4.22.4-12.el9_7        baseos         173 k
 samba-common-libs      x86_64      4.22.4-12.el9_7        baseos         104 k
 sssd                   x86_64      2.9.7-4.el9_7.1        baseos          25 k
 sssd-ad                x86_64      2.9.7-4.el9_7.1        baseos         217 k
 sssd-client            x86_64      2.9.7-4.el9_7.1        baseos         158 k
 sssd-common            x86_64      2.9.7-4.el9_7.1        baseos         1.6 M
 sssd-common-pac        x86_64      2.9.7-4.el9_7.1        baseos          94 k
 sssd-ipa               x86_64      2.9.7-4.el9_7.1        baseos         286 k
 sssd-kcm               x86_64      2.9.7-4.el9_7.1        baseos         107 k
 sssd-krb5              x86_64      2.9.7-4.el9_7.1        baseos          70 k
```


root@client:~
root@client:~
vagrant@client:~
<?xml version="1.0" encoding="utf-8"?>
<service>
<short>Samba Client</short>
<description>This option allows you to access Windows file and printer sharing networks.
You need the samba-client package installed for this option to be useful.</description>
<include service="netbios-ns"/>
<port protocol="udp" port="138"/>
</service>
/usr/lib/firewalld/services/samba-client.xml (END)

Open smb.conf /etc/samba Save

```
# See smb.conf.example for a more detailed config file or
# read the smb.conf manpage.
# Run 'testparm' to verify the config is correct after
# you modified it.
#
# Note:
# SMB1 is disabled by default. This means clients without support for SMB2 or
# SMB3 are no longer able to connect to smbd (by default).

[global]
        workgroup = USER-NET
        security = user

        passdb backend = tdbsam

        printing = cups
        printcap name = cups
        load printers = yes
        cups options = raw

[homes]
        comment = Home Directories
        valid users = %S, %D%w%S
        browseable = No
        read only = No
        inherit acls = Yes

[printers]
        comment = All Printers
        path = /var/tmp
        printable = Yes
        create mask = 0600
        browseable = No

[print$]
        comment = Printer Drivers
        path = /var/lib/samba/drivers
        # printadmin is a local group
```

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Activities smbusers (/etc/samba) - gedit

Open *smbusers /etc/samba Save

```
username=frank
password=houston237
```

```
# units generated from this file.
#
UUID=c84cce45-9089-48d9-9617-2f1d1bd45fdd /                       xfs     defaults        0 0
/swapfile none swap defaults 0 0
#VAGRANT-BEGIN
# The contents below are automatically generated by Vagrant. Do not modify.
vagrant /vagrant vboxsf uid=1000,gid=1000,_netdev 0 0
#VAGRANT-END

/server/sambashare /mnt/samba cifs
user,rw,uid=user_name,gid=sambagroup,
credentials=/etc/samba/smbusers,_netdev 0 0
```

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```
Added user user.
[root@server sambashare]# cd /vagrant/provision/server
[root@server server]# mkdir -p /vagrant/provision/server/smb/etc/samba
[root@server server]# cp -R /etc/samba/smb.conf /vagrant/provision/server/smb/etc/samba
/
[root@server server]# cd /vagrant/provision/server
[root@server server]# touch smb.sh
[root@server server]# chmod +x smb.sh
[root@server server]# gedit smb
smb/    smb.sh
[root@server server]# gedit smb.sh

(gedit:9421): dconf-WARNING **: 16:21:35.722: failed to commit changes to dconf: Failed
 to execute child process “dbus-launch” (No such file or directory)
```

smb.sh
/vagrant/provision/server

```
#!/bin/bash
LOGIN=frank
PASS=houston237

echo "Provisioning script $0"
echo "Install needed packages"
dnf -y install samba samba-client cifs-utils
echo "Copy configuration files"
cp -R /vagrant/provision/server/smb/etc/* /etc
chown -R root:root /etc/samba/*
restorecon -vR /etc
echo "Configure firewall"
firewall-cmd --add-service samba --permanent
firewall-cmd --reload
echo "Users and groups"
groupadd -g 1010 sambagroup
usermod -aG sambagroup $LOGIN
echo -ne "$PASS\n$PASS\n" | smbpasswd -L -a -s $LOGIN
echo "Make share dir"
mkdir -p /srv/sambashare
chgrp sambagroup /srv/sambashare
chmod g=rwx /srv/sambashare
echo "Tuning SELinux"
semanage fcontext -a -t samba_share_t "/srv/sambashare(/.*)?"
setsebool samba_export_all_rw 1
setsebool samba_export_all_rw 1 -P
restorecon -vR /srv/sambashare
echo "Start smb service"
systemctl enable smb
systemctl start smb
systemctl restart firewalld
```

Open smb.sh /vagrant/provision/server Save

```
1 #!/bin/bash
2 LOGIN=frank
3 PASS=houston237
4
5 echo "Provisioning script $0"
6 echo "Install needed packages"
7 dnf -y install samba samba-client cifs-utils
8 echo "Copy configuration files"
9 cp -R /vagrant/provision/server/smb/etc/* /etc
10 chown -R root:root /etc/samba/*
11 restorecon -vR /etc
12 echo "Configure firewall"
13 firewall-cmd --add-service samba --permanent
14 firewall-cmd --reload
15 echo "Users and groups"
16 groupadd -g 1010 sambagroup
17 usermod -aG sambagroup $LOGIN
18 echo -ne "$PASS\n$PASS\n" | smbpasswd -L -a -s $LOGIN
19 echo "Make share dir"
20 mkdir -p /srv/sambashare
21 chgrp sambagroup /srv/sambashare
22 chmod g=rwx /srv/sambashare
23 echo "Tuning SELinux"
24 semanage fcontext -a -t samba_share_t "/srv/sambashare(/.*)?"
25 setsebool samba_export_all_rw 1
26 setsebool samba_export_all_rw 1 -P
27 restorecon -vR /srv/sambashare
28 echo "Start smb service"
29 systemctl enable smb
30 systemctl start smb
31 systemctl restart firewalld
```

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```
(gedit:4874): dconf-WARNING **: 16:21:57.020: failed to commit changes to dconf: Failed to execute chi
ld process “dbus-launch” (No such file or directory)
[root@client samba]# mount -a
mount: /etc/fstab: parse error at line 20 -- ignored
mount.cifs: bad UNC (/server/sambashare)
mount: (hint) your fstab has been modified, but systemd still uses
       the old version; use 'systemctl daemon-reload' to reload.
mount: 0: mount point does not exist.
[root@client samba]# cd /vagrant/provision/client
[root@client client]# mkdir -p /vagrant/provision/client/smb/etc/samba
[root@client client]# cp -R /etc/samba/smb.conf /vagrant/provision/client/smb/etc/samba/
[root@client client]# cp -R /etc/samba/smbusers /vagrant/provision/client/smb/etc/samba/
[root@client client]# cd /vagrant/provision/client
[root@client client]# touch smb.sh
[root@client client]# chmod +x smb.sh
[root@client client]# gedit smb.sh

(gedit:4959): dconf-WARNING **: 16:32:55.868: failed to commit changes to dconf: Failed to execute chi
ld process “dbus-launch” (No such file or directory)

(gedit:4959): dconf-WARNING **: 16:32:55.874: failed to commit changes to dconf: Failed to execute chi
ld process “dbus-launch” (No such file or directory)

(gedit:4959): dconf-WARNING **: 16:32:56.084: failed to commit changes to dconf: Failed to execute chi
ld process “dbus-launch” (No such file or directory)

(gedit:4959): dconf-WARNING **: 16:32:56.110: failed to commit changes to dconf: Failed to execute chi
ld process “dbus-launch” (No such file or directory)

(gedit:4959): dconf-WARNING **: 16:32:56.110: failed to commit changes to dconf: Failed to execute chi
ld process “dbus-launch” (No such file or directory)
```

Open ▾ *smb.sh /vagrant/provision/client Save

Could not save the file "/vagrant/provision/client/smb.sh".

Unexpected error: Error renaming temporary file: Text file busy

```
#!/bin/bash
LOGIN=user
echo "Provisioning script $0"

mkdir -p /mnt/samba
echo "Install needed packages"
dnf -y install samba-client cifs-utils
echo "Copy configuration files"
cp -R /vagrant/provision/client/smb/etc/* /etc
chown -R root:root /etc/samba/*
restorecon -vR /etc
echo "Configure firewall"
firewall-cmd --add-service samba-client --permanent
firewall-cmd --reload
echo "Users and groups"
groupadd -g 1010 sambagroup
usermod -aG sambagroup $LOGIN
echo "Mounting dirs"
mkdir -p /srv/sambashare
echo "//server/sambashare /mnt/samba cifs
user,rw,credentials=/etc/samba/smbusers,uid=user,
gid=sambagroup,_netdev 0 0" >> /etc/fstab
↪
↪
restorecon -vR /etc
umount /mnt/samba
mount /mnt/samba
```

sh ▾ Tab Width: 8 ▾ Ln 24, Col 2 ▾ INS

Open *smb.sh /vagrant/provision/client Save

Could not save the file "/vagrant/provision/client/smb.sh".

Unexpected error: Error renaming temporary file: Text file busy

```
1 #!/bin/bash
2 LOGIN=user
3 echo "Provisioning script $0"
4
5 mkdir -p /mnt/samba
6 echo "Install needed packages"
7 dnf -y install samba-client cifs-utils
8 echo "Copy configuration files"
9 cp -R /vagrant/provision/client/smb/etc/* /etc
10 chown -R root:root /etc/samba/*
11 restorecon -vR /etc
12 echo "Configure firewall"
13 firewall-cmd --add-service samba-client --permanent
14 firewall-cmd --reload
15 echo "Users and groups"
16 groupadd -g 1010 sambagroup
17 usermod -aG sambagroup $LOGIN
18 echo "Mounting dirs"
19 mkdir -p /srv/sambashare
20 echo "//server/sambashare /mnt/samba cifs
21 user,rw,credentials=/etc/samba/smbusers,uid=user,
22 gid=sambagroup,_netdev 0 0" >> /etc/fstab
23 ↪
24 ↪
25 restorecon -vR /etc
26 umount /mnt/samba
27 mount /mnt/samba
```

sh Tab Width: 8 Ln 24, Col 2 INS

Вывод:

В ходе выполнения лабораторной работы были приобретены навыки настройки доступа групп пользователей к общим ресурсам по протоколу SMB.

Ответы на контрольные вопросы:

1. Какова минимальная конфигурация для smb.conf для создания общего ресурса, который предоставляет доступ к каталогу /data? -

   [global]

   workgroup = WORKGROUP

   security = user

   [data]

   path = /data

   read only = yes

2. Как настроить общий ресурс, который даёт доступ на запись всем пользователям, имеющим права на запись в файловой системе Linux? –

   [shared]

   path = /path/to/shared

   read only = no

3. Как ограничить доступ на запись к ресурсу только членам определённой группы? –

   [restricted]

   path = /path/to/restricted

   read only = no

   valid users = @group_name

4. Какой переключатель SELinux нужно использовать, чтобы позволить пользователям получать доступ к домашним каталогам на сервере через SMB? **- setsebool -P samba_enable_home_dirs on**
5. Как ограничить доступ к определённому ресурсу только узлам из сети 192.168.10.0/24? –

   [limited_access]

   path = /path/to/limited_access

   read only = yes

   valid users = @group_name

   hosts allow = 192.168.10.

6. Какую команду можно использовать, чтобы отобразить список всех пользователей Samba на сервере? **- pdbedit -L -v**
7. Что нужно сделать пользователю для доступа к ресурсу, который настроен как многопользовательский ресурс? –

   Иметь учетную запись на сервере.

   Быть добавленным в Samba с помощью smbpasswd -a username.

8. Как установить общий ресурс Samba в качестве многопользовательской учётной записи, где пользователь alice используется как минимальная учётная запись пользователя? –

   [alice_share]

   path = /path/to/alice_share

   read only = no

valid users = alice

9. Как можно запретить пользователям просматривать учётные данные монтирования Samba в файле /etc/fstab? - **Добавьте опцию credentials=/path/to/credentials_file в строке монтирования в /etc/fstab и сохраните файл с учетными данными за пределами публичного доступа.**
10. Какая команда позволяет перечислить все экспортируемые ресурсы Samba, доступные на определённом сервере? **- smbclient -L server_name**